Straight-line graphs — Plotting · Gradient · $y = mx + c$

Squared paper needed. ★ = challenge question.

1 Draw axes from -6 to 6. Draw the graph of $y = 2x - 3$ using a table of values for $x = -1, 0, 1, 2, 3$.

> $x = -1, 0, 1, 2, 3 \to y = -5, -3, -1, 1, 3$. Plot and join with a ruler: a straight line crossing the y-axis at -3, going up 2 for every 1 across.

2 Draw the graph of $3x + 2y = 6$.

> Easiest: $x = 0 \Rightarrow y = 3$; $y = 0 \Rightarrow x = 2$. Join $(0, 3)$ and $(2, 0)$. A third check point: $x = 4 \Rightarrow 2y = -6 \Rightarrow y = -3$.

3 Four lines: $y = x$, $y = 3x$, $y = 2x + 4$, $y = 2x - 1$.

- **a** Which lines pass through the origin?
- **b** Which line is the steepest?
- **c** Which two lines are parallel?
- **d** Which line crosses the y-axis lowest down?

> **a** $y = x$ and $y = 3x$ (no "$+c$", so $c = 0$). **b** $y = 3x$ — largest gradient. **c** $y = 2x + 4$ and $y = 2x - 1$ — same gradient 2. **d** $y = 2x - 1$, at $(0, -1)$.

4 Work out the gradient of the line through each pair of points.

a $(0, 1)$ and $(2, 5)$ **b** $(-2, 3)$ and $(4, 0)$ **c** $(1, -4)$ and $(3, -4)$

> Gradient $= \dfrac{\text{change in } y}{\text{change in } x}$. **a** $\dfrac{4}{2} = 2$ **b** $\dfrac{-3}{6} = -\dfrac{1}{2}$ **c** $\dfrac{0}{2} = 0$ — a horizontal line.

5 A line passes through $(0, -2)$ and $(4, 6)$. Write down its equation.

> Gradient $= \dfrac{6 - (-2)}{4 - 0} = 2$; it crosses the y-axis at -2. Equation: $y = 2x - 2$.

6

- **a** Where does the line $y = 3x - 5$ cross the y-axis?
- **b** What is the gradient of the line $y = -2x + 7$?
- **c** Write the equation of the line with gradient 4 that crosses the y-axis at $(0, -3)$.

> **a** At $(0, -5)$. **b** -2 (falling). **c** $y = 4x - 3$.

7 Does the point $(6, 2)$ lie on the line $y = 2x - 14$? Show working to explain.

> Put $x = 6$ into the equation: $y = 12 - 14 = -2$, not 2. So $(6, 2)$ is **not** on the line; $(6, -2)$ is.

8 Write the equation of each line.

- **a** Through $(0, 3)$ and $(3, 0)$.
- **b** Through $(0, 2)$ and $(1, 5)$.
- **c** Through $(0, -3)$ and $(4, -1)$.

> **a** Gradient $\dfrac{0-3}{3-0} = -1$, intercept 3: $y = -x + 3$. **b** Gradient 3, intercept 2: $y = 3x + 2$.
> **c** Gradient $\dfrac{2}{4} = \frac{1}{2}$, intercept -3: $y = \frac{1}{2}x - 3$.

9 A line $y = mx + c$ passes through the point $(1, 3)$. Write down three possible equations for the line. What is the relationship between m and c?

> Substituting $x = 1$, $y = 3$: $3 = m + c$. Any pair with $m + c = 3$ works: $y = x + 2$, $y = 2x + 1$, $y = -x + 4$, … — all these lines pass through $(1, 3)$.

10 Line A is $y = 2x + 1$. Line B is parallel to A and passes through $(3, 2)$. Line C is $y = -\frac{1}{2}x + 1$.

- **a** Find the equation of line B.
- **b** Where do lines A and C cross?

> **a** Same gradient 2: $y = 2x + c$; $2 = 6 + c \Rightarrow c = -4$. $y = 2x - 4$.
> **b** Both have intercept 1, so they cross on the y-axis at $(0, 1)$. (Algebra: $2x + 1 = -\frac{1}{2}x + 1 \Rightarrow x = 0$.)

Check: $y = mx + c$ — m is the gradient, c where it crosses the y-axis · parallel lines have the same m · a point is on the line only if its coordinates satisfy the equation.